LAB 3 - ARCHITECTURE JUSTIFICATION

Smart Lab Equipment & Slot Reservation Portal

Architecture Selection

We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal.

Architectural Choice

The system is divided into Presentation, Business, and Data layers. The Presentation layer handles student and lab-technician interactions. The Business layer contains authentication, equipment management, reservation management, and availability services. The Data layer stores equipment and reservation information.

Reason 1

Layered architecture provides clear separation of concerns. User-interface operations remain separate from reservation and equipment-management logic, making the portal easier to understand, test, and maintain.

Reason 2

The portal has a focused workflow: viewing availability, making or cancelling reservations, updating equipment status, and managing reservations. A layered design is simpler and more suitable than the operational complexity of microservices.

Security Advantage

Role-based access can be enforced by the Authentication & RBAC Service. Students can be restricted to student functions, while lab technicians can access equipment and reservation-management operations.

Performance Benefit

The design keeps frequently used availability and reservation operations close to the application logic and database, supporting the requirement to display availability and reservation information within 3 seconds for at least 95 out of 100 requests.